

Project Design Phase

Problem – Solution Fit

Date	05-07-2026
Team ID	SWTID-2026-8108
Project Name	Heart Disease Analysis
Maximum Marks	

Problem – Solution Fit :

- The Solution Fit: The Tableau dashboards. It translates 4,416 rows of messy data into a single, clean visual dashboard that answers complex questions instantly with 1-click filters.

Purpose:

Specifically, it bridges the gap between raw data and three distinct user perspectives to achieve the following:

- For Clinicians (Dr. Sharma): It allows them to instantly isolate and cross-filter high-risk patient groups (combining metrics like age, BMI, and smoking habits) to design targeted preventive medical campaigns.
- For Public Health Analysts (Ramesh): It maps lifestyle data against disease rates to compare urban vs. rural populations, giving them clear visual evidence to back up health policies and regulations.
- For Individual Patients (Anita): It translates messy health statistics into simple, color-coded benchmark comparison charts so individuals can track their personal health progress and reduce long-term risks.
- project takes 4,416 rows of complex, relational clinical data stored in MySQL and transforms it into an interactive Tableau dashboard that makes health risks instantly scannable, understandable, and actionable

Define CS, fit into CC	1. CUSTOMER SEGMENT(S) CS - Clinicians (Dr. Sharma) - Public Health Planners (Ramesh) - Proactive Patients (Anita)	6. CUSTOMER CONSTRAINTS CC - Lack of complex SQL or coding knowledge - Tight public health resource budgets - Anxiety over complex medical terminology	5. AVAILABLE SOLUTIONS AS - Raw static tabular spreadsheets - Traditional text-heavy medical files - Manual column filters in Excel	Explore AS, differentiate
	2. JOBS-TO-BE-DONE / PROBLEMS J&P - Spotting multi-attribute patient risk combinations quickly - Justifying healthcare budgets with regional metric data - Translating numeric clinical data into clear health insights	9. PROBLEM ROOT CAUSE RC - Raw clinical records are stored in complex, unstructured databases - Crucial risk metrics like BMI tiers and age distributions are hidden	7. BEHAVIOUR BE - Clinicians waste time running slow manual database searches - Policy makers guess regional trends due to poor visuals - Patients look up confusing medical stats online	Focus on J&P, map into BE, understand RC
	3. TRIGGERS TR - Rising regional heart disease cases - Missing hidden patient lifestyle risk trends during checkups	10. YOUR SOLUTION SL - Interactive MySQL-to-Tableau dashboard - Custom calculated risk categories (BMI and age groups) 1-click cross-filtering and professional natural language charts	8. CHANNELS of BEHAVIOUR CH 8.1 ONLINE: Interactive Tableau Public cloud link dashboards 8.2 OFFLINE: Local offline MySQL Workbench server queries & printable PDF summary reports	Extract online & offline CH of BE
Identify strong TR & EM				
4. EMOTIONS: BEFORE / AFTER EM - Before: Overwhelmed, frustrated, and anxious - After: Confident, fast-acting, and clear-minded				

